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Areal, Modular, and Connectional Organization of Visual Cortex in a Prosimian Primate, the Slow Loris (*Nycticebus coucang*)

Key Words

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Striate cortex
Extrastriate cortex
Cytochrome oxidase
Middle temporal area (MT)
Dorsolateral area (DL)
Strepsirhines
Prosimians
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Abstract

Slow lorises (*Nycticebus coucang*) are nocturnal prosimian (i.e. strepsirhine) primates, closely related to bushbabies (*Galago* spp.). We examined the organization of visual cortex in four hemispheres from two slow lorises, using connectional and architectonic techniques. All hemispheres were flattened and sections stained for myelin and cytochrome oxidase (CO). Our results indicate, first, that the primary visual area (V1) in slow lorises has a system of small CO-dense blobs, as has been described in most other anthropoid and prosimian primates examined to date. The second visual area (V2) is characterized by broad, stripe-like zones of dense CO staining separated by zones of lighter staining. Loris V2 stripes are less distinct than those of anthropoid primates, and separate classes of thin and thick dark stripes are not apparent. However, V2 stripes are much better developed than in *Galago*, where they are virtually absent. Injections of wheat-germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) in area V1 revealed reciprocal connections with area V2, and the middle temporal (MT) and dorsolateral (DL) extrastriate areas. Area MT was also identified by its distinctive, dense myelination. As has been reported in anthropoids, DL can be divided into separate caudal and rostral divisions, which differ in myelin and CO staining, and in the strength of their connections with V1. Taken together, our results suggest that many of the features that characterize visual cortex organization in anthropoid primates are present in prosimians and thus probably evolved early in primate history, prior to the diversification of modern primate groups.

Introduction

Mammalian cerebral cortex exhibits significant phyletic variation in virtually all aspects of its organization [Kaas, 1987; Preuss, 1993]. However, cortical variation is rarely the subject of systematic, comparative investigation. The lack of comprehensive comparative investigation makes it

difficult to develop and test detailed hypotheses regarding the evolutionary history of specific cortical systems. This is the case even for relatively well-studied systems and taxa. Visual cortex has been more intensively studied than any other part of the brain in primates, yet studies have been restricted to a remarkably few taxa, mainly three genera of anthropoid primates (*Macaca*, *Aotus*, *Saimiri*) and a single

Abbreviations

Aud	primary auditory cortex and neighboring auditory fields
CaS	calcarine sulcus
CgS	cingulate sulcus
CO	cytochrome oxidase
CS	central sulcus
DL	dorsolateral visual region
DLc	caudal division of the dorsolateral visual region
DLr	rostral division of the dorsolateral visual region
FST	fundal superior temporal visual area
HM	horizontal meridian
IAD	inferior arcuate dimple
IPS	intraparietal sulcus
IT	inferotemporal visual region
LS	lateral sulcus
LuS	lunate sulcus
M1	primary motor cortex
MT	middle temporal visual area
OB	olfactory bulb
OS	orbital sulcus
PcS	paracalcarine sulcus
PMV	ventral premotor area
S1	primary somatosensory cortex
STS	superior temporal sulcus
TMB	tetramethyl benzidine
V1	primary visual area
V2	second visual area
V4	fourth visual area
WGA-HRP	wheat-germ agglutinin conjugated to horseradish peroxidase

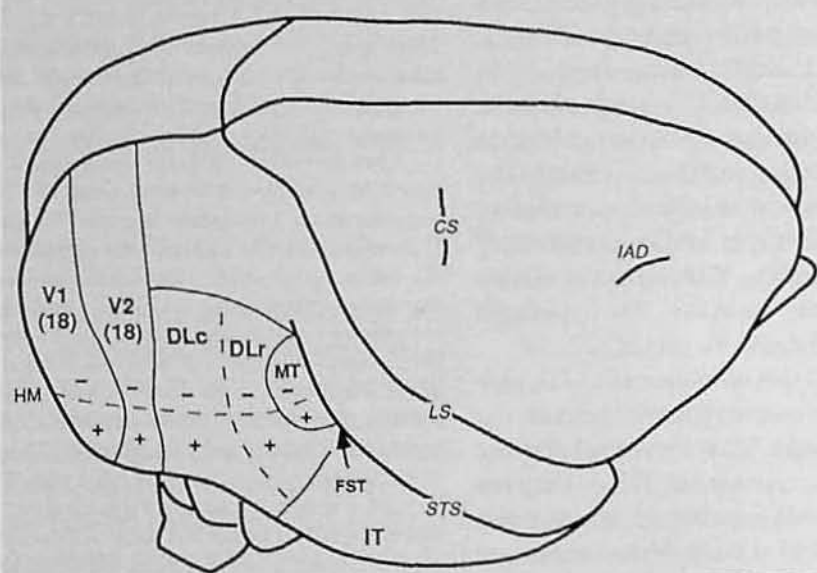
genus of prosimians (*Galago*). These taxa represent but a very small fraction of the diversity of living primates, which are comprised of approximately 60 genera and nearly 200 species [Fleagle, 1988]. Despite the limited comparative scope of research – which reflects in part the limited number of primate taxa available for research – hypotheses concerning possible shared, derived features of primate visual cortex have been advanced. For example, it has been suggested that certain extrastriate visual fields are common to all primates [Krubitzer and Kaas, 1990a; Kaas and Krubitzer, 1991]. Furthermore, it has been argued that some of these fields, for example the middle temporal visual area (MT), are absent in other mammals [Kaas and Preuss, 1993]. However, while extrastriate cortex has been intensively studied in a variety of anthropoid primates, among prosimians, only *Galago* has been studied in detail and demonstrated by multiple anatomical criteria to possess area MT [e.g. Allman et al., 1973; Symonds and Kaas, 1978]. Little is known about other extrastriate areas in prosimians. The differences in our current understanding of

visual cortex organization in anthropoids and prosimians are summarized in figure 1.

Evolutionary hypotheses have also been advanced regarding the subunits or 'modules' which make up primate visual areas. In the primary visual area (V1), the metabolic enzyme cytochrome oxidase (CO) is typically concentrated in an array of small puffs or 'blobs', first described in an unpublished letter by M. Wong-Riley, as discussed by Livingstone and Hubel [1984]. The cortex in blobs is connectionally and physiologically distinct from the 'interblob' regions of V1, and is thought to be part of a specialized color-processing channel in V1 [Livingstone and Hubel, 1988]. The V1 blobs have been observed in many primate taxa, and are absent in all non-primate mammals which have been examined [Horton and Hubel, 1981; Horton, 1984]. On this basis, Horton and Hubel have suggested that V1 blobs are unique to primates. However, McGuinness et al. [1986] have reported that blobs are absent in some nocturnal prosimian taxa, although they are present in at least one nocturnal prosimian, *Galago* [Horton, 1984; McGuinness et al., 1986; Condo and Casagrande, 1990; Krubitzer and Kaas, 1990a]. The second visual area (V2) also shows

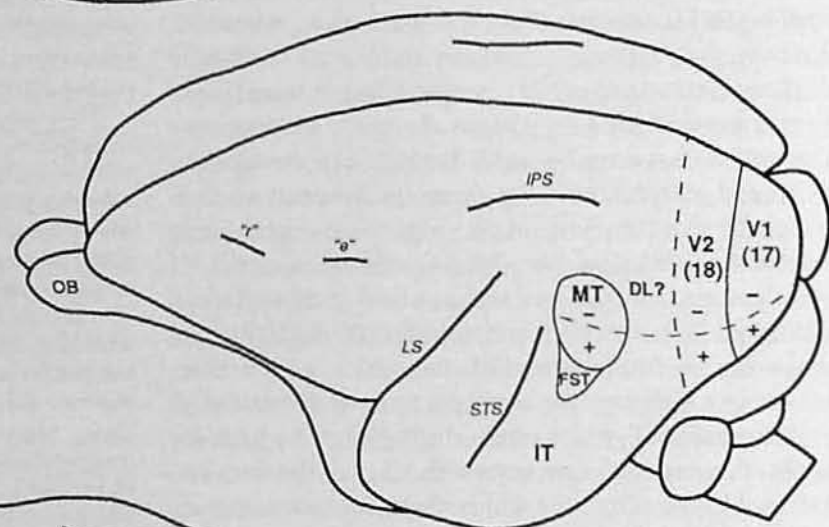
Fig. 1. Cortical organization in *Saimiri*, *Galago*, and *Nycticebus*. The squirrel monkey (*Saimiri*), a New World anthropoid primate, has been studied intensively and much is known about the areal and retinotopic organization of its visual cortex. This figure is based on Cusick and Kaas [1988a] and Steele et al. [1991]. V1, V2, and MT are areas known in a variety of primate taxa. The area lying between V2 and MT is termed the dorsolateral (DL) region in New World monkeys and the V4 region in Old World monkeys. DL is currently divided into caudal (DLc) and rostral (DLr) areas, based on differences in architectonics and connections. In all these visual areas, the lower visual field (–) is represented dorsally and the upper visual field (+) ventrally; the horizontal meridian of the visual field (represented as a dashed line) thus lies on the lateral surface of the hemisphere, bisecting areas V1, V2, DLc, DLr, and MT. The representation of the central visual field is located rostrally in V1, in the adjacent, narrow part of V2, and in the caudal part of MT. *Saimiri* possesses additional visual areas, including FST, located in the fundus of the superior temporal sulcus (STS), and the inferotemporal visual cortex (IT).

Compared to anthropoid primates, prosimian primates have received little study; the best-known taxon is *Galago*. V1, V2, and MT have been documented in *Galago*, and there is limited evidence for DL, FST, and IT [see especially Cusick and Kaas, 1988b; Krubitzer and Kaas, 1990a, and Preuss and Goldman-Rakic, 1991a, b]. Subdivisions of DL have not been described, however. *Nycticebus* visual cortex has not been studied by modern workers. Note the convoluted appearance of *Nycticebus* cortex, which has numerous sulci, including a shallow lunate sulcus (LuS) in the occipital lobe. The features marked 'c', 'e', and 'r' are shallow grooves present in the parietal and frontal lobes of some prosimians [Haines et al., 1974]. Additional abbreviations: CS – central sulcus; IAD – inferior arcuate dimple; IPS – intraparietal sulcus; LS – lateral sulcus; OB – olfactory bulb.

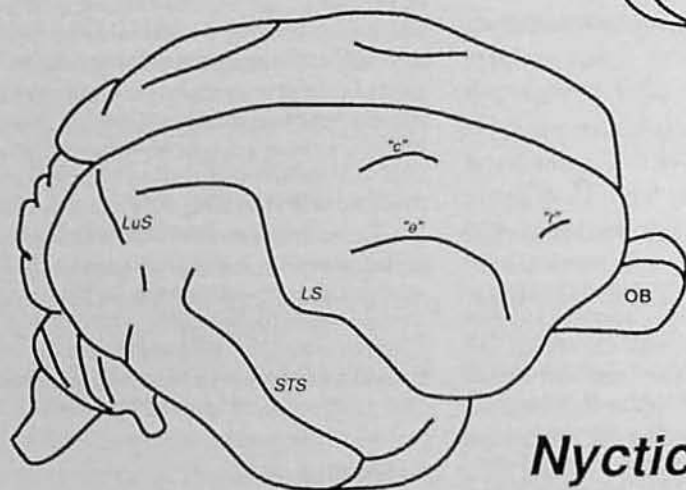


Saimiri

5 mm



Galago



Nycticebus

periodic variation in CO staining. In all anthropoid primates studied, V2 exhibits a striking pattern of CO-dense bands or stripes, alternating with pale, lightly stained bands [e.g. Tootell et al., 1983; Wong-Riley and Carroll, 1984; Hubel and Livingstone, 1985; Krubitzer and Kaas, 1989, 1990a, b]. By contrast, in *Galago* V2, CO staining is more homogeneous, and distinct stripes are not apparent (Condo and Casagrande, 1990; Krubitzer and Kaas, 1990a). The absence of distinct V2 stripes in the prosimian *Galago* raises the possibility that V2 stripes are an evolutionary innovation of anthropoid primates. Without information about V2 in other prosimians, however, this hypothesis lacks a solid empirical foundation.

We report here an investigation of visual cortex in a prosimian that has not previously been examined in detail, the slow loris (*Nycticebus coucang*). Slow lorises inhabit the tropical forests of southeast Asia [Bearder, 1986]. They are closely related to galagos, both taxa being placed in the family Lorisidae [Fleagle, 1988]. Like galagos, lorises are small-bodied (averaging about 650 grams) and nocturnal. However, their locomotor anatomy and behavior, adapted for slow climbing and crypsis, is very different from that of the active, quick-moving galagos. In order to determine whether lorises possess V1 and V2 stripes, we examined the striate and extrastriate visual cortex in flattened sections stained for cytochrome oxidase. Aspects of connectional and areal organization were investigated by injecting V1 with a sensitive bidirectional tracer, wheat-germ agglutinin conjugated to horseradish peroxidase (WGA-HRP), and comparing the distribution of labeling with respect to myelin and CO staining. We were particularly interested in localizing area MT, which can be distinguished by its dense myelination and its connections with V1, and the dorsolateral visual area (DL), for which there is at present only limited evidence in prosimians.

Materials and Methods

Investigations were carried out in two adult male slow lorises (*Nycticebus coucang*). Both animals were old, with heavily worn teeth and gingival trauma. When they became unable to maintain themselves, it was decided to euthanize them. The animals were anesthetized with intramuscular injections of ketamine hydrochloride (30 mg/kg) and xylazine (1–2 mg/kg). Supplemental doses of these drugs were administered as needed throughout the surgical procedure to maintain a deep level of anesthesia. A small opening was made in the cranium and dura over the right occipital lobe. Injections of wheat-germ agglutinin conjugated to horseradish peroxidase (WGA-HRP) were then placed in the exposed, dorsolateral surface of the cortex, within presumptive area V1. In case 91-86R, a total of 0.15 μ l of 1% WGA-HRP was injected in a single penetration. In 91-90R, a total of 0.08 μ l of 2%

WGA-HRP was injected in two closely spaced injections. After surgery, animals were allowed to recover from the anesthesia and were administered analgesics and antibiotics. Following survival periods of two days (91-86R) or four days (91-90R), each animal was given a lethal dose of barbiturate and perfused intracardially with buffered saline, followed by 2% paraformaldehyde, and finally by a solution of 2% paraformaldehyde and 10% sucrose. All procedures were carried out under guidelines established by the National Institutes of Health.

After perfusion, the brain was removed from the skull and prepared for flattening. The left frontal lobe was separated from the remainder of the hemisphere and used for the study of motor cortex [Preuss et al., 1992]. The posterior cortex of the left hemisphere and the entire cortex of the right hemisphere were separated from the thalamus by blunt dissection of the white matter. In order to unfold the primary visual cortex, much of which is buried within the calcarine sulcus along the medial wall, a cut was made along the depth of the calcarine sulcus as far as the occipital pole, prior to flattening. A second cut was made through the rostral part of the lateral sulcus to help unfold the temporal and parietal lobes. The unfolded cortical pieces were then placed in a Petri dish filled with 30% sucrose, and pressed beneath a weighted glass cover slip for about 12 hours. The flattened blocks were then frozen and sectioned parallel to the cortical surface at 40 μ m. A series of 1:3 sections was processed with tetramethyl benzidine (TMB) to reveal HRP, using a modification of the technique of Gibson et al. [1984]. Alternate sections were stained for cytochrome oxidase [Wong-Riley, 1979] or myelin [Gallyas, 1979].

Histological sections were examined with darkfield and brightfield illumination, and drawings of HRP injection sites and HRP-labeled cells, axons, and terminals were made at 10 $\times$ using a microscope with a drawing tube. The drawings were subsequently checked for accuracy and completeness at higher magnification. In sections processed for TMB, the location of WGA-HRP injections were marked by a zone of very dense, diffuse reaction product, appearing opaque under darkfield illumination, surrounded by a halo of lighter diffuse reaction product. Following Mesulam [1982], we considered the zone of dense reaction product to constitute the region of uptake and transport. To illustrate the distribution of labeling with respect to architectonic areas, images of CO- and myelin-stained sections were projected onto the drawings of HRP labeling. Additional drawings were made of CO- and myelin-stained sections to locate architectonic borders in the uninjected left hemispheres and to measure the dimensions of areas, blobs, and stripes. We examined only flattened sections of *Nycticebus* visual cortex in the present study. As a general rule, the laminar pattern of staining cannot be determined precisely using flattened sections. However, the internal granular layer (layer 4) of area V1 has been found to stain very densely and uniformly for CO in previous primate studies [Horton, 1984], and using this feature as a marker of layer 4 in *Nycticebus*, we were able to determine whether sections passed through the supragranular or infragranular layers. The analysis of laminar and areal patterns of staining was further complicated by the difficulty encountered in flattening *Nycticebus* cortex (especially the large right-hemisphere blocks), resulting from the presence of many small sulcal grooves and very tough, adherent pia mater.

Results

Using flattened sections stained for CO and myelin, we were able to identify several areal divisions of visual cortex

in *Nycticebus*. Based on comparisons of staining patterns in anthropoid primates, areas V1, V2, MT, and DL were identified in *Nycticebus*. In CO-reacted sections, V1 was characterized by the presence of blobs and V2 by the presence of irregular patches and stripes. The distribution of cortical labeling after V1 injections was consistent with our interpretation of *Nycticebus* areal organization, which is summarized in figure 2.

Appearance of Visual Cortex in CO- and Myelin-Stained Sections

CO staining of V1 and V2. The primary visual area was readily recognized at the occipital pole of the hemisphere in CO-stained sections. Sections taken through middle levels of cortex exhibited uniform and very dark staining, similar to the pattern of staining seen in layer 4 in other primates, as discussed above. Also, as shown in figure 3, sections taken from levels superficial to layer 4 revealed a regular array of small, darkly stained patches or blobs. Blob-like periodicities were also evident at levels deep to layer 4. Whereas in some primates, such as macaques, blobs are aligned into columns [Horton, 1984], no regular alignment of blobs was apparent in *Nycticebus*. CO blobs were typically slightly elongated or elliptical in appearance. Measurements of blob dimensions were taken from sections through the supragranular layers in the rostral part of V1, corresponding to the central visual field representation. These measurements yielded median values of 0.2×0.27 mm (based on a sample of 46 blobs from 3 hemispheres). The median center-to-center distance between blobs was 0.5 mm ($n=57$ intervals from 3 hemispheres).

Through the superficial layers of cortex, the border between V1 and V2 was marked by the replacement of small CO-dense blobs with larger CO-dense patches (fig. 3). In many instances, neighboring patches merged into stripes, oriented approximately perpendicular to the V1-V2 border. The patches and stripes varied considerably in thickness, ranging from 0.3–1.1 mm, with a median thickness of 0.55 mm ($n=12$ stripes from 3 hemispheres). This variability suggests that *Nycticebus* may possess two size classes of stripes, comparable to the alternating thin and thick stripes seen in some anthropoids [Tootell et al., 1983; Krubitzer and Kaas, 1989]. However, *Nycticebus* stripes were very irregular in appearance, and no alternating pattern of thin and thick stripes could be discerned with this material. The irregularities of the stripes also made it very difficult to accurately measure the distance between stripes. However, in two hemispheres, we measured center-to-center interstripe intervals ranging from 1.35–3.0 mm, with a median of 1.7 mm ($n=7$ intervals).

The rostral border of V2 was marked by the disappearance of the dark CO patches and stripes. The width of V2 varied across the visual field representation. It was most narrow on the lateral surface of the hemisphere, in the region just ventral to the lunate sulcus (LuS). At this point, V2 extended approximately 2.5–3.5 mm rostrocaudally. It broadened at more ventrolateral and dorsomedial levels to a width of approximately 4–5 mm. The narrowing of V2 in *Nycticebus* is similar to that described for the central visual field representation of V2 in *Saimiri* and other primates [Krubitzer and Kaas, 1990a; Steele et al., 1991], shown in figure 1.

Myelin Staining of V1 and V2. Area V1 could also be distinguished in myelin-stained sections. In sections through the middle levels of cortex, V1 exhibited a system of faint blob-like patches, similar to the CO blobs in more superficial sections (fig. 4). By contrast, area V2 did not show a distinctive pattern of myelin staining, and the rostral border of V2 was not clearly marked in myelin-stained sections. The V1-V2 border was marked by a distinct, myelin-sparse gap in the superficial and middle cortical layers. This gap may reflect the incomplete unfolding of shallow grooves and small sulci in these cases. The border between V1 and V2 was located in close proximity to the LuS on the dorsal surface of the occipital lobe. In two hemispheres (91-90R and L) the border ran through the sulcus, while in two others (91-86R and L) the border lay somewhat caudal to the sulcus.

MT and DL. The middle temporal visual area was identifiable in sections through middle levels of cortex as a discrete, round or slightly ovoid territory of very dense myelination, with a rostrocaudal width of approximately 3–4 mm (fig. 4). Area MT also stained rather darkly for CO in our *Nycticebus* material but was much less easily distinguished from neighboring areas in CO sections than in myelin-stained sections. Area MT was situated rostral to the central visual field representations of V1 and V2, in close approximation to the dorsal end of the superior temporal sulcus (STS). In three hemispheres, the STS passed along the caudal border of MT, forming a limiting sulcus (fig. 4, 5), although in one hemisphere (91-90R) the STS ended within the caudal and ventral part of the area, and the folding of cortex around the end of the STS probably obscured the ventral border of the area (fig. 6). In some sections, a small, triangular zone of heavy myelination could be seen extending from MT ventrally along the STS, which probably corresponds to area FST (fig. 4). This area (named for the 'fundus of the superior temporal sulcus') was originally described in macaque monkeys [Desimone and Ungerleider, 1986], and similar zones have been described in other primates (fig. 1).

Nycticebus coucang

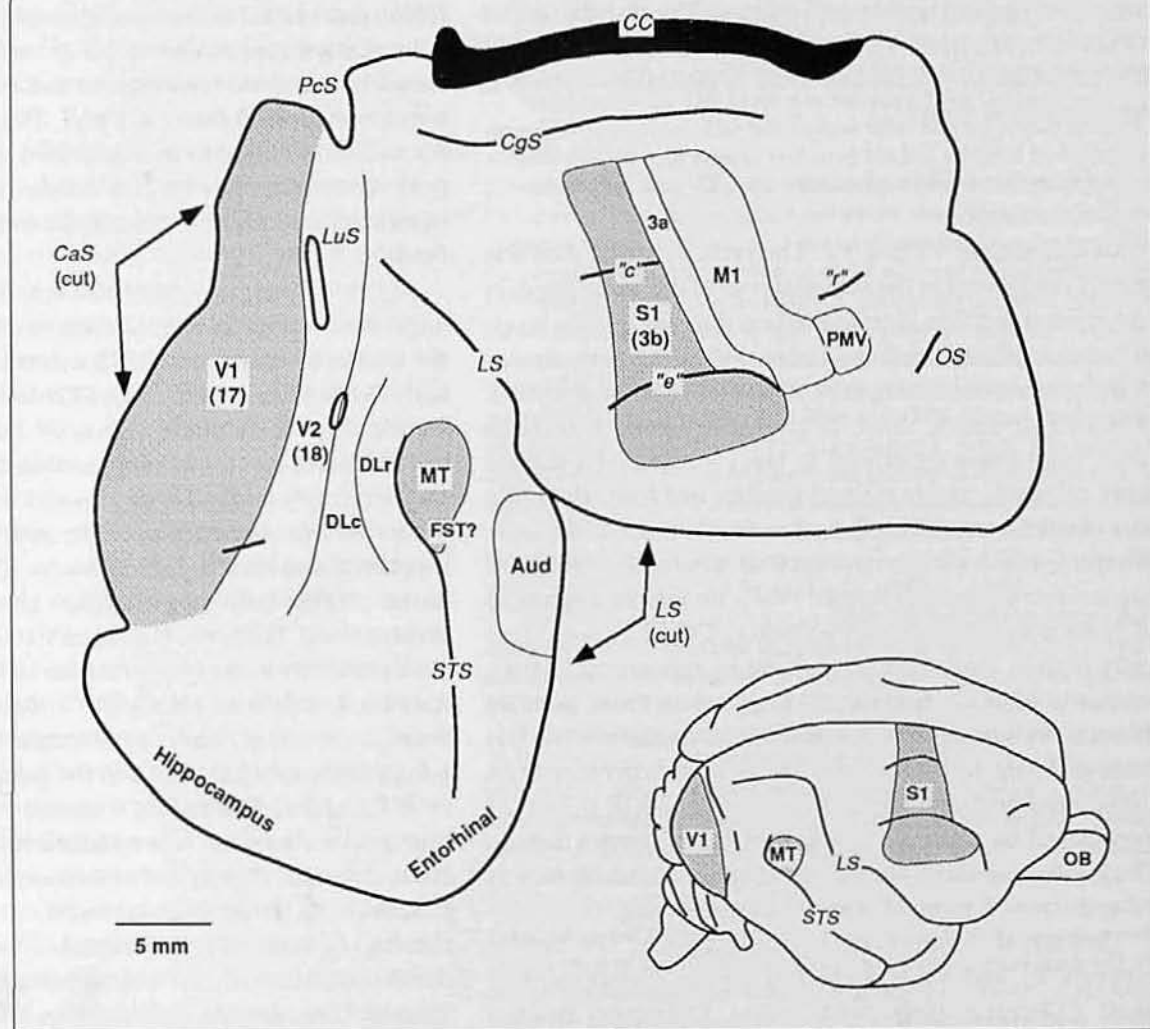


Fig. 2. A schematic, unfolded representation of cortical organization in *Nycticebus*, summarizing our architectonic parcellation of visual cortex using flattened sections stained for myelin and cytochrome oxidase. In order to flatten the cortex, cuts were made along the fundus of the calcarine sulcus (CaS) and lateral sulcus (LS). Note the narrowing of V2 caudal to DL and MT; this probably corresponds to the location of the horizontal meridian and the central visual field representation, as illustrated for *Saimiri* in figure 1. Also depicted are the primary somatosensory (S1) cortex and area 3a [Krishnamurti et al., 1976], and the primary motor (M1) and ventral premotor (PMV) areas [Preuss et al., 1992]. Additional abbreviations: Aud – primary auditory cortex and neighboring auditory fields; CC – corpus callosum; CgS – cingulate sulcus; OS – orbital sulcus; PcS – paracalcarine sulcus.

The region of cortex interposed between MT and V2 has been termed the dorsolateral visual area (DL) in New World monkeys [Allman and Kaas, 1974a]. In squirrel monkeys, Cusick and Kaas [1988a] have divided the DL region into separate caudal and rostral areas (DLc, DLr) which differ in myeloarchitecture and CO staining. Our

architectonic observations in *Nycticebus* are consistent with results in *Saimiri*. In myelin-stained sections, DLc formed a crescent of moderately dense myelination partly enveloping the more lightly myelinated DLr (fig. 4). Differences were also visible in CO-stained sections through the deep cortical layers, where DLc were more heavily stained than

Fig. 3. A flattened section through the visual cortex in a left hemisphere, case 91-90L, stained for cytochrome oxidase. Caudal is to the left, dorsal to the top. The section passes through cortex superficial to layer 4. In V1, small CO-dense blobs (indicated by small arrows) stand out against a background of more lightly stained tissue. Large CO-dense puffs and stripes (indicated by small stars) are apparent in V2. The curved arrow marks a small sulcus located along the rostral border of V2. Abbreviations are the same as in figures 1 and 2. Scale bar=2 mm.

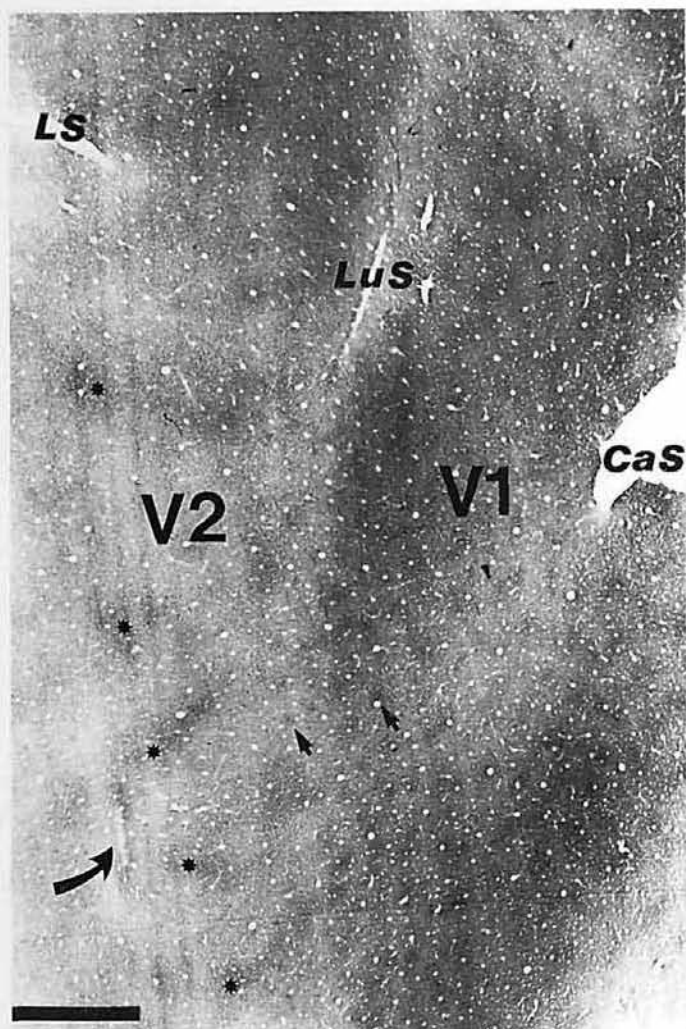
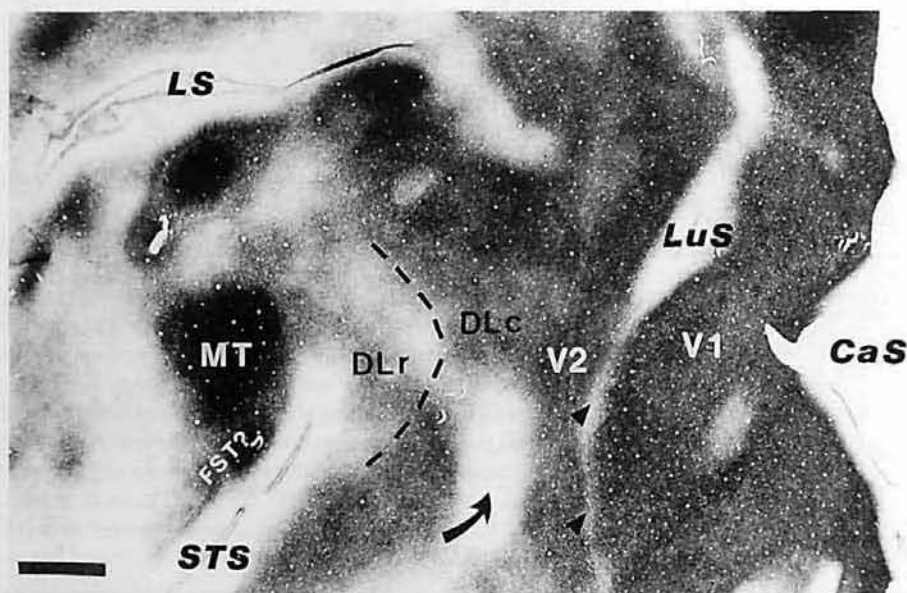


Fig. 4. Flattened section through the left visual cortex (case 91-90L) stained for myelin. The orientation of this section is similar to that shown in figure 3, although the magnification is lower and a larger region of occipital and temporal cortex is visible. This section illustrates the dense myelination of area MT, and the distinction between the pale-staining DLr and more darkly staining DLc. A narrow, myelin-sparse zone (denoted with arrowheads) marks the border between V1 and V2. The curved arrow marks the small sulcus located along the border between V2 and DLc. Abbreviations are the same as in figures 1 and 2. Scale bar=2 mm.



91-86R *Nycticebus*

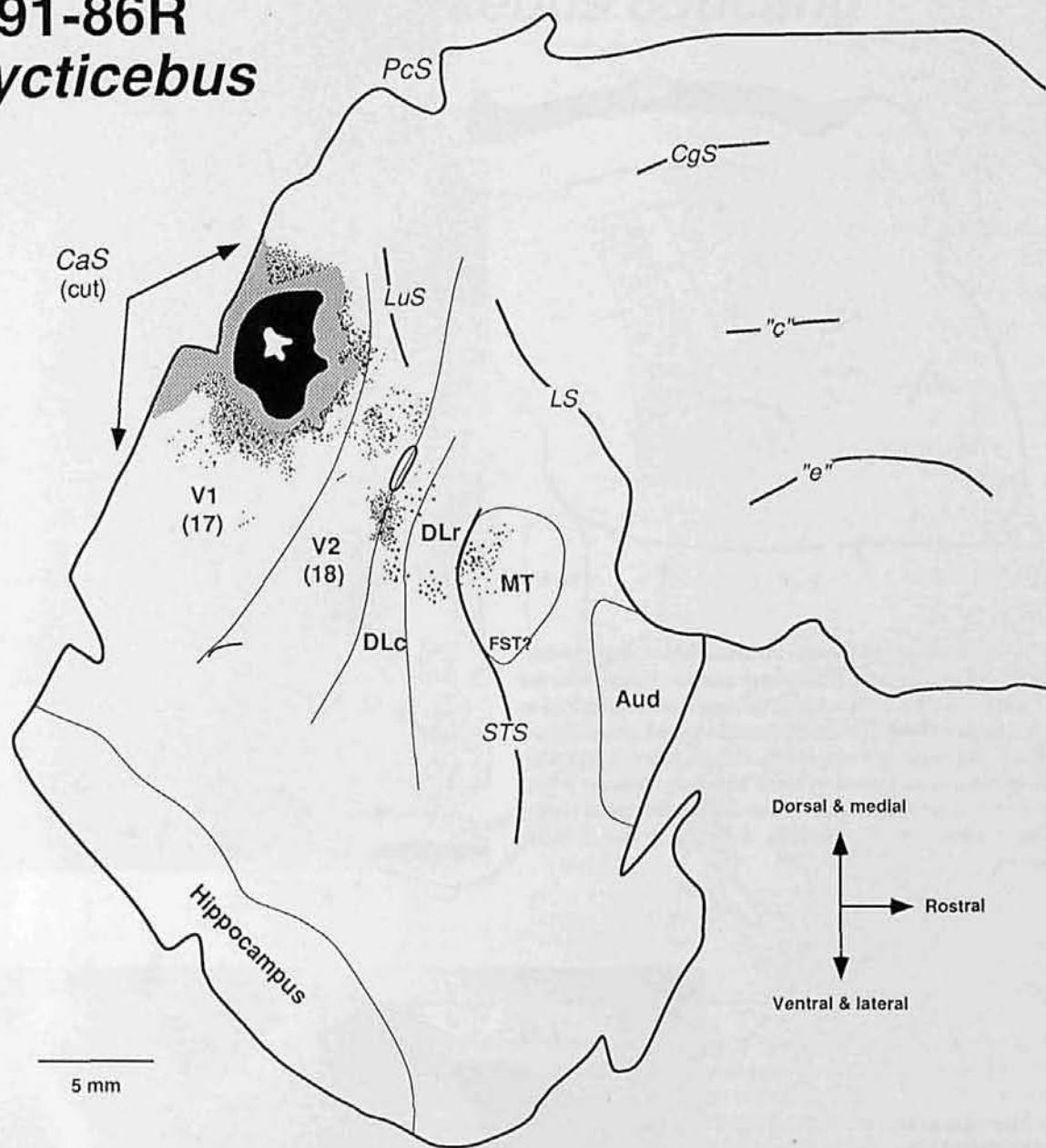


Fig. 5. Ipsilateral cortical labeling after a V1 injection of WGA-HRP in case 91-86R. The injection site was located in the presumptive central representation of V1, close to the border with V2. Large dots represent retrogradely labeled cells; small dots represent anterogradely labeled fibers and terminals. Areal borders were drawn from myelin- and CO-stained sections. Abbreviations are the same as in figures 1 and 2.

Fig. 6. Ipsilateral cortical labeling after V1 injections of WGA-HRP in case 91-90R. The injections in this case were placed further from the V1-V2 border than in 91-86R, and thus involved somewhat more of the peripheral visual field representation. The injection probably also involved more of the lower field representation, as suggested by the concentration of label in dorsomedial V2, rostral to the LuS. The ventrolateral border of MT is dashed to indicate that this area probably extended further along the STS than was apparent in myelin-stained sections. Large dots represent retrogradely labeled cells; small dots represent anterogradely labeled fibers and terminals. Abbreviations are the same as in figures 1 and 2.

91-90R *Nycticebus*

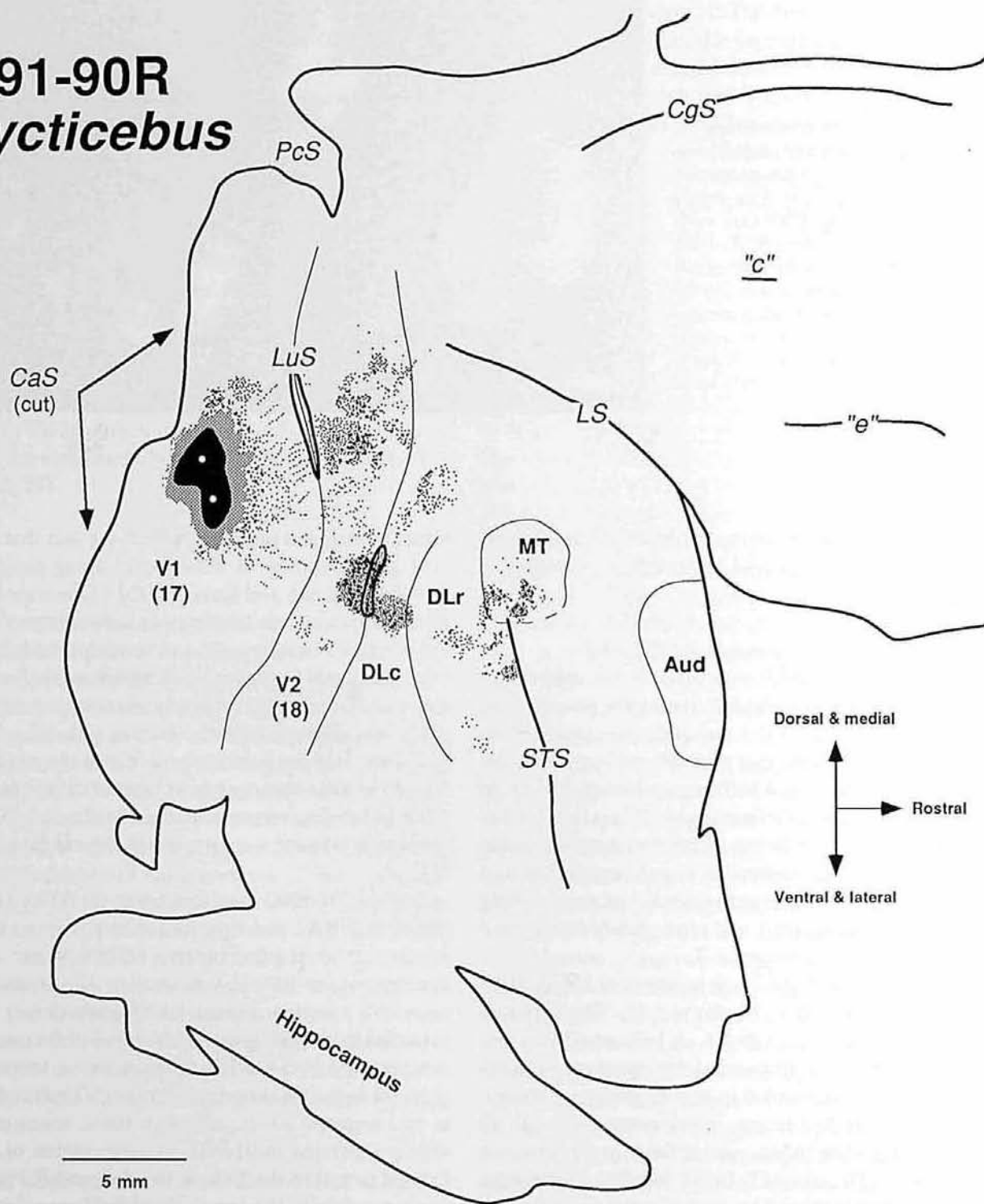
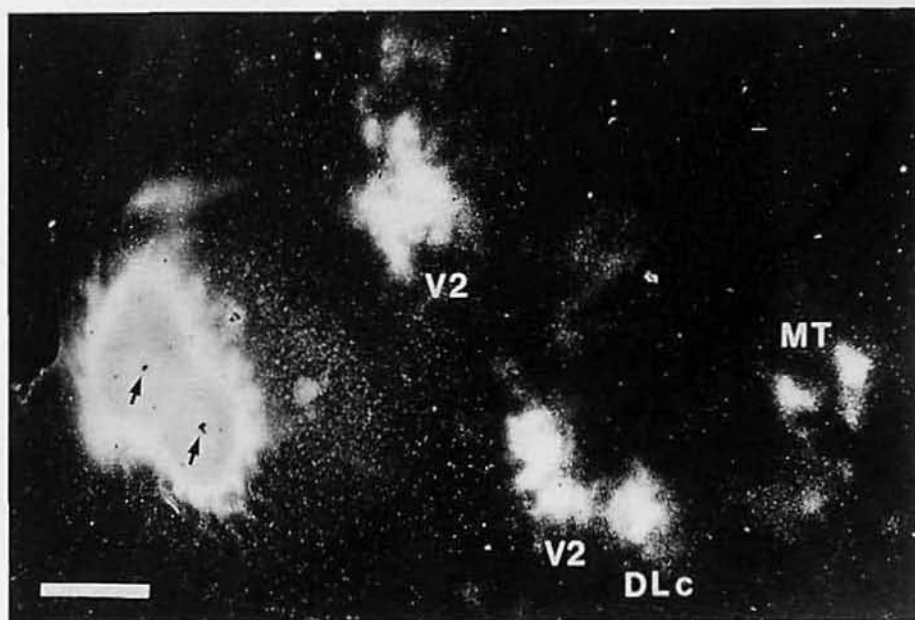


Fig. 7. Darkfield photomicrograph of the injection site and ipsilateral cortical labeling in case 91-90R. Compare this micrograph with the drawing in figure 6. Two closely spaced injections of WGA-HRP were made in this case; the needle tracks are marked with arrows. Two patches of dense anterograde and retrograde label can be seen in V2; the upper patch is in the lower field representation and the lower patch is in the central representation. Dense label is also present in DLc. Several patches of dense label can be seen in MT. Scale bar=2 mm.



DLr. A small dimple marked the border between DLc and V2 in the region of the central visual field representation (fig. 2, 4-6).

Ipsilateral Cortical Connections of Area V1

Injections of WGA-HRP were made in the right occipital cortex in two animals. The injections were placed in the portion of V1 located on the dorsolateral surface of the occipital lobe, in locations that probably correspond to the central or paracentral visual field representation (fig. 1). In both cases, transported label was present in areas V2, DLc, DLr, and MT. The distribution of label within these areas was consistent with the central or paracentral location of injection sites in V1. The injection sites, and the resulting distribution of anterogradely and retrogradely transported label, are illustrated in figures 5-7.

In case 91-86R, a single, large injection of WGA-HRP was placed in V1 close to its border with V2. The apparent injection site was restricted to V1. A halo of diffuse, less dense reaction product surrounded the injection site proper. The halo was surrounded in turn by a zone of anterograde and retrograde labeling, representing intrinsic V1 connections. Labeling of extrastriate fields in this case was restricted to V2, DL, and MT. In V2, labeling was concentrated in two closely spaced territories in the central visual field representation of the area. (The occurrence of two foci of labeling in V2 after a single injection of V1 is consistent with the previous primate studies [e.g. Cusick and

Kaas, 1988a], and probably reflects the fact that the visual field representation of V2 is 'split' along the horizontal meridian [Allman and Kaas, 1974b].) One zone of anterograde and retrograde labeling was located immediately rostral to the V1 injection site and ventrolateral to the lunate sulcus. A second, smaller patch of dense label was located more laterally in the relatively narrow part of V2. This patch was situated near the border with DL, and dense label extended for a short distance into the caudal part of DL. Area DLr also contained labeled cells, but fiber and terminal labeling was very sparse. Finally, anterograde and retrograde labeling were present in the caudal part of area MT.

In case 91-90R, two injections of WGA-HRP were placed within V1 and appeared to merge into a single, relatively restricted injection site. Although the combined injection site in 91-90R was smaller than in the previous case, with a single injection, labeling was denser and more extensive in 91-90R, presumably owing to the use of a more concentrated WGA-HRP solution and a longer survival time. As in the previous case, label in V2 was concentrated in two separate zones, although these zones were more widely separated in 91-90R. A large region of label was located rostral to the LuS, in the dorsomedial part of V2, corresponding to the lower visual field representation. A second patch of dense label was located more laterally in V2, in the central visual field representation. This focus involved the rostral part of V2 and extended into DLc.

Additional light label was observed medial to the dense focus in DLc. This would be consistent with labeling of a lower field representation in DLc, comparable to the lower field labeling of V2, although we were not able to confirm architectonically that the label was within DLc. Immediately rostral to the labeled focus in DLc, light label was present in DLr. As in 91-86R, fiber and terminal labeling was rather sparse compared to the density of cell labeling in DLr. Ventrolateral to the main labeled region of DLr, there were two small patches of anterograde label that may have been located within DLr. The most rostral labeling in 91-90R was located around the dorsomedial end of the STS, in the vicinity of area MT. Three small concentrations of anterograde and retrograde labeling were definitely located within the densely myelinated cortex of MT. Labeling continued ventrolaterally along the STS. Some of this may have been within MT, although, as discussed above, the full ventrolateral extent of MT in this case was probably obscured by the folding of cortex around the STS. It seems likely that the most laterally situated labeling along the STS fell outside MT.

Discussion

The present study is one of the very few investigations of visual cortex in a prosimian primate other than *Galago*. With regard to the modular (or subareal) organization of visual cortex, *Nycticebus* resembles anthropoids and *Galago* in possessing an array of CO-dense blobs in V1. In V2, *Nycticebus* possesses irregular stripe-like forms which stain densely for CO. The V2 stripes are better developed in *Nycticebus* than in *Galago*, but are less regular and distinct than those of anthropoid primates. The areal and connectional organization of visual cortex in *Nycticebus* resembles that reported for *Galago* and for anthropoid primates in most respects. Specifically, we were able to identify extrastriate areas V2, DLc, DLr, and MT in *Nycticebus*, and to demonstrate that these areas are interconnected with V1. The comparative evidence suggests that many basic features of the areal and connectional organization of visual cortex were established early in primate evolution and were inherited without extensive modification by living primate taxa.

Modular Organization of Slow Loris Visual Cortex: V1 Blobs and V2 'Stripes'

In Old World and New World anthropoid primates, and in the prosimian primate *Galago*, workers have demonstrated the existence in V1 of a periodic pattern of small

CO-dense puffs or blobs [e.g. Tootell et al., 1983; Horton, 1984; Hess and Edwards, 1987]. Our CO material indicates that *Nycticebus* also possesses V1 blobs. Blobs were visible at two levels in the cortex. The most well-marked periodicities were observed in the supragranular layers, although periodicities were also evident in the deepest level of cortex. This is consistent with published descriptions of the laminar distribution of CO staining in anthropoids and in *Galago* [Horton, 1984; Condo and Casagrande, 1990]. Area V1 blobs vary in size across taxa: as a rule, species with larger cortices have larger blobs [Horton, 1984]. For example, blobs are larger in the greater bushbaby, *Galago crassicaudatus*, than in the smaller Senegal or lesser bushbaby, *Galago senegalensis* [Condo and Casagrande, 1990]; humans have the largest blobs yet described [Horton and Hedley-Whyte, 1984]. Our quantitative results indicate that the V1 blobs of *Nycticebus coucang* are similar in size to those of *Galago crassicaudatus*, an animal that is similar in body size and brain size to *N. coucang* [Martin, 1990]. However, the statistical relationship between blob size and cortical surface area may not be very strong, as blob size in *Nycticebus* is similar to that reported for *Macaca* [Horton, 1984], which has a much larger brain.

Area V1 blobs are thus present in at least two genera of prosimian primates, *Nycticebus* and *Galago*, in addition to New World and Old World anthropoid primates. Because they are present in both major subdivisions of the primate order, prosimians and anthropoids, and are absent in all other mammals that have been examined, Horton's [1984] proposal that V1 blobs are a 'unique, defining' feature of primate neocortex is well founded [see also Kaas and Preuss, 1993]. Although V1 blobs are probably *derived* features of primate organization, it is nevertheless possible that they are not *universally* present in primates. For instance, McGuinness et al. [1986] have reported that blobs are absent in three prosimian taxa, *Cheirogaleus*, *Hapalemur*, and *Tarsius*. This report must be considered with some caution, however, as the tissue used in that study came from animals that died some time before the brains could be fixed and processed for CO. Although the authors make a reasonable case that their negative results were not artifacts of poor fixation and delayed processing, it is important that these results be confirmed in animals that are perfusion-fixed at the time of death.

The existence of blobs in primate V1 is a matter of considerable functional interest. In macaques and squirrel monkeys, many neurons located within V1 blobs are color selective, and on this basis it has been suggested that blobs represent a specialized color channel in V1 [Livingstone and Hubel, 1988]. However, as pointed out by Horton

[1984], Allman and Zucker [1990], and Condo and Casagrande [1990], the generality of this conclusion is challenged by the fact that blobs are well developed in at least some nocturnal primates (*Nycticebus*, *Galago*, and *Aotus*), and studies of nocturnal primates indicate that they have little color vision [Jacobs, 1977; Pariente, 1979; Condo and Casagrande, 1990]. Moreover, the earliest primates were almost certainly nocturnal [Martin, 1990]. Therefore, it seems unlikely that blobs originated in evolution as adaptations for color vision. Clues to the original function of blobs may be provided by the properties of the non-color selective cells present in blobs, which consist of units sensitive to variations in luminance [Livingstone and Hubel, 1988]. Allman and Zucker [1990] suggest that blobs may have evolved as part of a system for maintaining perceptual brightness constancy in the face of variations in illumination.

In contrast to the dot-like pattern of CO staining observed in V1, area V2 contains a system of broad CO-dense stripes, oriented perpendicular to the V1-V2 border, and separated by lightly stained zones. This pattern has now been reported in a variety of anthropoid taxa, including *Macaca*, *Papio*, *Homo*, *Saimiri*, *Callithrix*, *Cebus*, and *Aotus* [Tootell et al., 1983; Horton, 1984; Horton and Hedley-Whyte, 1984; Tootell et al., 1985; Hess and Edwards, 1987; Cusick and Kaas, 1988a; Krubitzer and Kaas, 1989, 1990a, b]. Some studies have reported that the system of V2 stripes is composed of alternating thin and thick CO-dense bands, which receive projections from the blob and interblob regions of V1, respectively [Tootell et al., 1983; Livingstone and Hubel, 1984]. Although prosimian V2 has not been extensively investigated, studies suggest that stripes are absent or at most very poorly developed in *Galago*, at least. Condo and Casagrande [1990] and Krubitzer and Kaas [1990a] reported that stripe-like CO periodicities were not apparent in V2 of *Galago senegalensis*, although they did observe some faint, periodic staining in *G. crassicaudatus*. Our own unpublished observations in *G. crassicaudatus* are in accord with these findings.

Compared to *Galago*, area V2 in *Nycticebus* has more distinct and darkly stained CO features. These features are quite irregular, in some instances appearing as large isolated patches, in others as elongated, stripe-like forms. However, in none of the *Nycticebus* hemispheres we studied did the stripes appear as regular or well formed as those depicted in published micrographs of anthropoid primates. Furthermore, we were not able to consistently distinguish thin stripes and thick stripes in *Nycticebus*.

The loridid prosimians *Nycticebus coucang*, *Galago crassicaudatus* and *G. senegalensis* thus exhibit considerable

interspecific variation in density and patterning of CO staining in V2. While anthropoids generally exhibit stronger staining than prosimians, variation has also been noted in these animals. For example, V2 stripes are darker and more distinct in some anthropoids, such as *Saimiri*, than in others, such as *Aotus* and *Callithrix* [Tootell et al., 1985; Krubitzer and Kaas, 1990a, b]. Furthermore, whereas thin stripes and thick stripes can be readily distinguished in *Saimiri*, the V2 stripes of *Callithrix* appear to be of a single thickness [Krubitzer and Kaas, 1990a].

What is the significance of the observed variability in CO staining of area V2 across primates? One possibility is that variations in CO staining reflect variations in the connectional organization of V2. In anthropoids, different classes of cortical inputs and outputs are segregated among the thin stripes, thick stripes, and pale stripes. In macaques, for example, the thick stripes receive inputs from layer 4B of area V1, while the thin stripes receive inputs from layer 4C; the thick stripes project in turn to area MT and the thin stripes to area V4 (DL) [Livingstone and Hubel, 1988]. Variation in CO staining might therefore reflect variations in the degree to which inputs and outputs are segregated in V2. Contrary to this proposal, however, Krubitzer and Kaas [1990a] have noted that in *Galago senegalensis*, as in anthropoids, projections to MT arise from repeated stripe-like arrays of cells in V2, even though these arrays do not stain differentially for CO. Thus the connectional organization of V2 may be very similar in prosimians and anthropoids, despite differences in the staining pattern for cytochrome oxidase. The reason that primates vary in the CO staining pattern of area V2 remains to be explained.

Architectonic and Connectional Identification of Areas MT, DLc, and DLr

Comparative anatomical evidence suggests that most mammals possess cortex homologous to anthropoid areas V1 and V2, and a small number of additional, higher-order visual areas [Kaas, 1989; Kaas and Krubitzer, 1991]. By contrast, anthropoid primates possess a very large number of higher-order visual areas, perhaps 20 or more [Felleman and Van Essen, 1991; Kaas and Krubitzer, 1991]. In order to better understand the evolutionary history of primate visual cortex, it is important to identify homologous areas in the two major groups of primates. Our results in *Nycticebus*, and the results of similar studies in *Galago*, suggest that loridid prosimians possess homologues of at least three of higher-order areas which have been identified in anthropoids, specifically MT, DLc, and DLr.

The middle temporal visual area (MT) has been identified in a variety of primates based on its myeloarchitec-

tonic, connectional, and physiological properties. As originally described in owl monkeys [Allman and Kaas, 1971], MT is an oval-shaped area in the temporal lobe, characterized by very dense myelination and a complete, topographically organized representation of the contralateral visual field. It was soon discovered that MT has strong, reciprocal connections with V1 [Spatz and Tigges, 1972; Tigges et al., 1973]. Based on these characteristics, MT has been identified in a variety of anthropoid primate taxa [Kaas and Krubitzer, 1991], but only in a single prosimian taxon, *Galago* [e.g. Allman et al., 1973; Tigges et al., 1973].

We were able to identify MT in *Nycticebus* as a small zone of heavy myelination, located along the rostral bank of the dorsal STS, which was well labeled after injections of V1. Its location in *Nycticebus*, rostral to the central visual field representations of V1 and V2, resembles that of other primates. Area MT in *Nycticebus* is very similar in size and shape to that in *Galago crassicaudatus* [Cusick and Kaas, 1988b; Preuss and Goldman-Rakic, 1991a]. However, the location of MT in *Nycticebus* – along the anterior bank of the STS – is unusual. In other primates with a distinct superior temporal sulcus, MT is located either posterior to the sulcus or buried (partly or completely) within the posterior bank of the sulcus. This is the case in *Galago* and *Saimiri*, as illustrated in figure 1, and in other New World and Old World monkeys [Kaas and Krubitzer, 1991].

The cortex interposed between MT and V2 on the lateral surface has been identified as the dorsolateral visual area (DL) in New World monkeys or as area V4 in Old World monkeys [Felleman and Van Essen, 1991; Kaas and Krubitzer, 1991]. Originally described in owl monkeys as a single, retinotopically organized area with an enlarged representation of central vision [Allman and Kaas, 1974a], current evidence indicates that the DL region is comprised of caudal (DLc) and rostral (DLr) divisions that differ architectonically and connectionally [Cusick and Kaas, 1988a; Steele et al., 1991; Weller et al., 1991]. In anthropoids, DLc has recently been shown to have connections with V1 [Steele et al., 1991]. These connections were not consistently demonstrated by earlier studies [reviewed by Weller and Kaas, 1981, and Van Essen, 1985]. Little is known about the corresponding region in prosimian primates. Early studies in prosimians did not reveal connections between V1 and DL [Symonds and Kaas, 1978; Tigges et al., 1981]. However, as in anthropoids, a later study using more sensitive techniques revealed connections between V1 and the cortex interposed between V2 and MT [Cusick and Kaas, 1988b], suggesting that prosimians possess cortex homologous to DL [see also Wall et al., 1982].

Evidence for separate rostral and caudal divisions of DL has not yet been supplied, however.

Our results suggest that the organization of the DL region in *Nycticebus* is very similar to that of anthropoids. We were able to distinguish caudal and rostral architectonic zones within this region. Area DLc was characterized by moderately dense staining of myelin and CO, while DLr was more lightly stained in both preparations. This is consistent with the architectonic appearance of DLc and DLr as described in recent studies of *Saimiri* [Cusick and Kaas, 1988a; Steele et al., 1991]. Moreover, we found that DLc has strong, reciprocal connections with V1, and our results suggest that DLc is visuotopically organized, with central vision represented immediately rostral to the central representation of V2 and the lower visual field represented more dorsomedially. In these respects, our connectional results in *Nycticebus* are in accord with those of Steele et al. [1991] for *Saimiri* DLc. The rostral division of DL is also connected with V1, although weakly. Injection of V1 in *Nycticebus* resulted in sparse labeling in DLr, with a high proportion of retrograde to anterograde labeling. This suggests that the projections from V1 to DLr are weaker than the reciprocal projections. The pattern of V1-DL connectivity documented here for *Nycticebus* is very similar to that recently described in *Galago* by Cusick and Kaas [1988b]. Although the DL region was not characterized architectonically in that study, injections of WGA-HRP in V1 yielded label in the region between V2 and MT that was relatively dense caudally and light rostrally. Collectively, these results provide evidence that at least some prosimian primates possess cortical divisions homologous to anthropoid areas DLc and DLr, and that these divisions evolved early in primate history. It should be noted that in anthropoid primates, connections between V1 and DLr have not been reported [e.g. Lin et al., 1982; Steele et al., 1991]. However, the connections of DLr in anthropoids have not yet been investigated in detail, and in view of the fact that V1-DLr connections are weak in *Nycticebus*, it would be premature to conclude that anthropoids and prosimians differ in this aspect of connectivity.

There is an additional difference between the connectional results reported here for *Nycticebus* and those reported for anthropoids. In anthropoids, V1 is known to be connected with the cortex located rostral and dorsomedial to V2, a region that has been termed the dorsomedial visual area (DM) in New World monkeys [Lin et al., 1982; Krubitzer and Kaas, 1990b; Weller et al., 1991] and dorsal V3 in Old World monkeys [Cragg, 1969; Zeki, 1980; Burkhalter et al., 1986]. Comparable connections are not demonstrated in most previous studies of prosimians. However,

Cusick and Kaas [1988b] illustrated two *Galago* cases in which light label was present in the dorsomedial region after a WGA-HRP injection in V1, suggesting that V1-DM connections are not an anthropoid specialization.

The comparative evidence above highlights the high degree of similarity in the areal and connectional organization between prosimian and anthropoid primates. It is possible to reconstruct a common primate 'plan' of organization, which includes the modular segregation of different classes of visual information in the blob and interblob regions of V1, and the subsequent transmission of this information from V1 to a common set of extrastriate visual areas, including V2, DL, and MT. This comparative anatomical interpretation has additional ramifications for understanding the evolution of visual processing in primates. In anthropoids, DL and MT represent early stages in a fundamental division of labor among higher-order visual areas, which have been termed the ventral and dorsal processing streams, or the 'what' and 'where' systems [Ungerleider and Mishkin, 1982]. Specifically, the divisions of DL project to the inferior temporal cortex, a region specialized for the visual discrimination of object form (the 'what' system), while MT projects to the posterior parietal cortex, which is specialized for spatial analysis (the 'where' system). The demonstration that MT, DLc, and DLr exist in prosimians, taken together with connectional and architectonic evidence that prosimians possess posterior parietal and inferotemporal cortex [Krubitzer and Kaas, 1990a; Preuss and Goldman-Rakic, 1991a, b], indicates that the

dorsal and ventral processing systems were probably present in the last common ancestor of the living prosimian and anthropoid primates. This said, we do not wish to discount the possibility that important differences may yet be discovered in the visual cortical regions of prosimians and anthropoids. It is important to bear in mind the extremely restricted phyletic scope of published prosimian studies: virtually all research to date has been carried out in species of the single family Lorisidae, while very little information is available for other major prosimian groups, including the diverse lemurs of Madagascar (comprising the families Lemnidae, Lepilemuridae, Indridae, Daubentoniidae, and Cheirogaleidae), or for *Tarsius*, the prosimian thought to be most closely related to anthropoids. Moreover, our understanding of the organization of extrastriate cortex in mammalian groups closely related to primates, the so-called 'archontan' mammals (tree shrews, bats, and colugos), is very rudimentary [Kaas and Preuss, 1993]. Investigations of these taxa may yet substantially modify our understanding the ancestral organization and subsequent evolution of visual cortex in primates.

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